

ECEN-662: Homework 2

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Problem 1

i) Let

$$f_{\theta}(x) = h(x) \exp(\eta^t T - A(\eta))$$

be an s -parameter exponential family of distributions. Show that if there exists

$$\{\theta_0, \theta_1, \dots, \theta_s\} \subseteq \Theta$$

such that the vectors

$$(\eta(\theta_i) - \eta(\theta_0) : i \in [s])$$

are linearly independent in $\mathcal{R}^s$, then T is a minimal sufficient statistic.

ii) Let $X = (X_1, \dots, X_n)$, where X_i 's are i.i.d. $\mathcal{N}(\theta, \theta^2)$, and $\theta > 0$ is the unknown parameter. Use the result of part i) to show that

$$T = \left(\sum_{i=1}^n X_i, \sum_{i=1}^n X_i^2 \right)$$

is a minimal sufficient statistic.

Solution

Part (i): Since the ratio $f_{\theta}(x)/f_{\theta}(y)$ is independent of θ , and it implies that $T(x) = T(y)$, T is a minimal sufficient statistic.

Part (ii): Here we have pdf

$$f_{\theta}(x) = \frac{1}{\sqrt{2\pi\theta^2}} \exp\left(-\frac{(x-\theta)^2}{2\theta^2}\right) = \frac{1}{\sqrt{2\pi\theta^2}} \exp\left(-\frac{x^2}{2\theta^2} + \frac{x}{\theta} - \frac{1}{2}\right)$$

so $T = (X, X^2)$ is a sufficient statistic.

Problem 2

Suppose that T_1 is sufficient and T_2 is minimally sufficient. Let U be an unbiased estimator of θ and define

$$U_1 = \mathbb{E}_\theta[U|T_1] \quad \text{and} \quad U_2 = \mathbb{E}_\theta[U|T_2]$$

Show that $U_2 = \mathbb{E}_\theta[U_1|T_2]$ and hence $\text{Var}_\theta[U_2] \leq \text{Var}_\theta[U_1]$.

Solution

Using the definition of U_2

$$U_2 = \mathbb{E}_\theta[U|T_2] = \mathbb{E}_\theta[\mathbb{E}_\theta[U|T_1]|T_2] = \mathbb{E}_\theta[U_1|T_2]$$

and for the variance

$$\text{Var}_\theta[U_1] = \text{Var}(\mathbb{E}_\theta[U_1|T_2]) + \mathbb{E}_\theta[\text{Var}_\theta[U_1|T_2]] = \text{Var}[U_2] + \mathbb{E}_\theta[\text{Var}_\theta[U_1|T_2]] \geq \text{Var}_\theta[U_2]$$

Problem 3

Let $X = (X_1, \dots, X_n)$, where X_i 's are i.i.d. $\mathcal{U}([0, \theta])$, and $\theta > 0$ is the unknown parameter. Suppose g is a differentiable function on $\mathcal{R}_{++}$. Find an UMVUE of $g(\theta)$ in terms of the function g .

Solution

Here the pdf is

$$f_{\theta}(t) = nt^{n-1}\theta^{-n}1_{\{0 \leq t \leq \theta\}}$$

and the expectation is

$$\mathbb{E}_{\theta}[T] = \int_0^{\theta} nt^n\theta^{-n}dt = \frac{n\theta}{n+1}$$

so then $\hat{\theta} = \frac{n+1}{n}T$ must be an UMVUE of θ .

Problem 4

Let $X = [X_1, \dots, X_n]^t$, where

$$X_i \stackrel{iid}{\sim} \text{Poisson}(\lambda),$$

and $\lambda > 0$ is the unknown parameter. Suppose we wish to estimate

$$\theta = \exp(-\lambda) = \mathbb{P}_\lambda(X_i = 0).$$

- i) Show that $T = \sum_{i=1}^n X_i$ is a complete sufficient statistic for this likelihood family.
- ii) Show that $S_1 = 1_{\{X_1=0\}}$ and $S_2 = \frac{1}{n} \sum_{i=1}^n 1_{\{X_i=0\}}$ are both unbiased estimate of θ .
- iii) Compute the Rao-Blackwellized estimators $\tilde{S}_i = \mathbb{E}_\theta[S_i|T]$, $i = 1, 2$. What does your answer have to do with the completeness of T ?

Solution

Part (i): Here the pdf is

$$f_\lambda(x) = \prod_{i=1}^n \frac{\lambda^{x_i} e^{-\lambda}}{x_i!} = \frac{\lambda^{\sum_{i=1}^n x_i} e^{-n\lambda}}{\prod_{i=1}^n x_i!}$$

so $T = \sum_{i=1}^n X_i$ is a sufficient statistic.

Part (ii): The expected value of S_1 is

$$\mathbb{E}_\lambda[S_1] = \mathbb{P}_\lambda(X_1 = 0) = \exp(-\lambda) = \theta$$

and the expected value of S_2 is

$$\mathbb{E}_\lambda[S_2] = \frac{1}{n} \sum_{i=1}^n \mathbb{P}_\lambda(X_i = 0) = \exp(-\lambda) = \theta$$

so both S_1 and S_2 are unbiased estimators of θ .

Part (iii): Here

$$\tilde{S}_1 = \mathbb{E}_\theta[S_1|T] = \mathbb{E}_\theta[S_2|T] = \tilde{S}_2$$

thus the estimators are equal.